

Fixed-height shifted rectangles

An unconditional proof of the A181199 asymptotic,
all-order expansions, and an algebraicity dichotomy

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For every fixed number of rows m , let $T_m(n)$ count the $m \times n$ arrays increasing in rows, columns, diagonals, and downward antidiagonals. The principal result is

$$T_m(n) \sim \frac{\sqrt{m} \prod_{j=0}^{m-1} j!}{2^{m(m-1)} (2\pi)^{(m-1)/2}} \frac{m^{mn}}{n^{(m^2-1)/2}}.$$

The proof supplies every inverse-power correction, proves the asymptotic conjecture in OEIS A181199, and implies that $\sum_{n \geq 0} T_m(n) z^n$ is algebraic if and only if $m \leq 2$.

Abstract. We split the order-polytope model of a shifted rectangle at a fixed numerical threshold. Except for an explicitly bounded exponentially small boundary contribution, its enumeration is an expectation under independent binomial random variables, weighted by a squared Vandermonde and a rational pair interaction. The shifted hook formula identifies the weight exactly. Standardized binomial moments are polynomials in $1/n$, so the expectation has an effective all-order expansion with rational coefficients. Its leading Gaussian integral gives the constant above; the first two corrections are proved as explicit polynomials in m . We obtain the conjectured constant $9\sqrt{5}/(2^{17}\pi^2)$ for A181199 and five correction terms, a universal limiting ratio to ordinary rectangular tableaux, nonalgebraicity at every height at least three, and Gaussian-ensemble laws for random intermediate row lengths. Complete proofs, an exact-arithmetic coefficient engine, independent enumerators, and a source/status audit are included.

Status. AI-assisted, unrefereed research manuscript. The theorems have proofs below, but have not been independently peer reviewed or proof-assistant verified. Historical priority is not certified. No conjectured recurrence is used as a premise, and none is claimed proved here.

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1 The problem, the open target, and the source audit

For positive integers m, n , consider arrays $(a_{i,j})$ containing each integer in $\{1, \dots, mn\}$ exactly once, subject to

$$a_{i,j} < a_{i,j+1}, \quad a_{i,j} < a_{i+1,j}, \quad (1.1)$$

$$a_{i,j} < a_{i+1,j+1}, \quad a_{i,j+1} < a_{i+1,j}, \quad (1.2)$$

whenever the indicated cells exist. Write $T_m(n)$ for their number, and set $T_m(0) = 1$. The diagonal inequality is redundant once row and column increase are imposed. The downward-antidiagonal inequality is not.

These numbers form the array A181196. Its fixed-height rows include A181197 for $m = 3$, A181198 for $m = 4$, and A181199 for $m = 5$ [1, 2, 3, 4]. Throughout this article, m is fixed and the row length n tends to infinity. Interchanging these two parameters changes the problem: already $T_2(n) = C_{n-1}$ for $n \geq 1$, whereas $T_m(2) = 1$ for every m .

1.1 The conjecture resolved here

The A181199 entry inspected on October 3, 2026, labels the following formula as a conjecture of Vaclav Kotesovec, dated February 27, 2023:

$$T_5(n) \sim \frac{9 \cdot 5^{n+1/2}}{2^{17} \pi^2 n^{12}}. \quad (1.3)$$

The entry says it was based on Christoph Koutschan’s conjectured recurrence. Kauers and Koutschan discuss the relevant guessed recurrences and finite-sum conjectures for heights four and five in their 2023 article [7, 4]. We do not assume those recurrences. We prove (1.3) directly from the combinatorial definition and extend it to every fixed height.

1.2 What is established background, and what is contributed

The shifted hook product and the order-statistics interpretation are classical tools, not new inventions of this report. Sun’s treatment of shifted strips gives both tools and proves enumeration results in several small cases [8]. The known height-three leading asymptotic in A181197 is recovered, not newly conjectured. Chan’s transfer-matrix theory concerns repeating strips with a fixed number of cells in each row; in the notation here that is the regime with n fixed and m growing [9]. Its constant-coefficient recurrences do not supply a fixed-height asymptotic in n .

The contributions developed in this manuscript are the exact random-threshold reduction with a boundary bound, the resulting all-height all-order theorem, universal first and second corrections, an algebraicity classification, and intermediate-shape limit laws. These statements are proved below. The source review did not locate these statements in the works inspected, but that observation is not an exhaustive priority search. In particular, a formula’s continued “conjecture” label in OEIS is not by itself proof that no published proof exists.

1.3 Relationship to ProveIt

The repository was inspected at revision 6bf7f30d0352f7596e70928b3d4f304914075907. Identifier searches for A181198 and A181199 returned no matching repository report. This is a limited search result, not a claim to have audited every repository file. The existing path-forest reports distinguish unconditional

expansions, conjectured recurrences, and inverse-index approximations [14]. We retain that separation. No theorem in that report is a mathematical dependency of the present proof, and the standard inverse-index method in Section 10 is not presented as new.

Scope of the resolution. Equation (1.3) is proved here. The order-two recurrence for A181198 and the order-three recurrence for A181199 are *not* proved here. The asymptotic series is an all-order Poincaré expansion, with a fixed truncation order; no Borel summability, Stokes data, or complete exponentially improved transseries is asserted.

[write] Added 3 October 2026, batch 85. This report is a single manuscript, printed in full: manuscript 09 of batch 85 of ProveIt’s incoming-reports intake, delivered as the archive `Shifted_Rectangle_Asymptotics.zip` (343,338 bytes, wrapper directory `Shifted_Rectangle_Asymptotics/`) in commit 9d6968c8a, and placed in commit ddf8df5d5 in the research-report collection, among the OEIS-sequence asymptotics reports, as the report `a181199-shifted-rectangles`. The title page reads “Prepared for Vladimir Reshetnikov”; the PDF metadata names “Research report prepared for Vladimir Reshetnikov”. No human author and no tool are named. The delivered text is printed unchanged except that every label carries the prefix `shr:`, the title page no longer creates a duplicate page anchor, and dated notes such as this one are added.

Pin and repository claims. The revision quoted above, 6bf7f30d0 (3 October 2026), is the ProveIt commit the manuscript was written against. At placement, and again at this writing, the intake searched the whole tracked tree for A181196–A181199, shifted rectangles, shifted tableaux, shifted hook products and Sun’s article; apart from the arrival archive, the only hits are the batch-85 sibling named below, which points to this report. This report continues no repository material, and the sentence “Identifier searches for A181198 and A181199 returned no matching repository report” remains true. “The existing path-forest reports” is one report, `oeis-sequence-asymptotics/a189281-path-forest-expansions`, in three Parts; as the manuscript says, none of its theorems is used here. No statement of this report is formalized in Lean or Rocq, no formal development in ProveIt treats shifted tableaux, and the report’s place in the collection confers no formal status.

Neighbours. The companion reports of the same batch are `a047874-long-increasing-subsequences` (manuscript 08), whose fixed-sector problem also uses the hook count of the rectangle with m rows of length n as a normalizer, and `a182220-source-boundary` (manuscript 07, unrelated mathematics). Other conjectures of the 2023 article of Kauers and Koutschan [7] are proved in the collection reports `enumerative-combinatorics/a181280-binary-matrix-formula` (its Conjecture 20, Section 6.5, a C-finite formula), `enumerative-combinatorics/a195806-hexagonal-lattice` (Conjecture 11) and Part IV of `a215561-fixed-composition-excursions` (Conjecture 15). The present report concerns that article’s Section 6.4, and leaves the guessed recurrences of that section unproved. Two notes below relate Corollary 2.4 and Section 10 to the transseries volumes in `Analysis/Transseries/docs/series-and-transseries/`.

Notation. The manuscript reuses several letters with local meanings. They are kept as delivered; the table fixes each reading. No symbol was renamed.

[write] Notation table of the note in Section 1.3 (added 3 October 2026)

Symbol	Meanings in this report
$T_m(n)$	the shifted-rectangle count (A181196 family), <i>not</i> the number $f^{(n^m)}$ of standard Young tableaux of the $m \times n$ rectangle; the transseries volume writes $T_d(n)$ for the latter, with d rows.
d	$\binom{m}{2}$, the number of row pairs ((2.1)); in the transseries volume d is the number of rows, m here.
$K_m; K; K_j(X)$	the leading constant (2.2); $K = K_m$ in Section 10, but the number of samples below p in the proof of Theorem 9.2; Krawtchouk polynomials in Section 6.1.
$L; L_5(n)$	a truncation order (Theorem 2.1); $\log(y/K)$ in Section 10; the leading term in Section 7.3.
$h_r(q); h_j$	complete homogeneous polynomials (5.3); squared norms $j! n^j/n^j$ in Section 6.1.
λ	a state or strict partition (Section 3, Appendix A); $m \log m$ in Section 10.
$R_n; R_m; R(z); \mathcal{R}_{L,n}$	the pair weight (4.2); the radius m^{-m} in Section 8; the rational function of Appendix A; the Taylor remainder (5.12).
$p; p_r$	the threshold in $(0, 1)$; the power sums $\sum_i Y_i^r$ (Section 6).
$q_{ij}; q$	$(y_i + y_j)^2$; a derivative order in Section 8.
$Z_m(n); Z_i$	the binomial Vandermonde normalizer (6.1); standardized counts in Section 9.
$A, B; \mathcal{A}$	$\binom{m}{3}$ and $m(m-1)^2(m-2)/12$ (Section 6.1); the event that all comparisons hold.
$C(z); C_{n-1}; C_{m,L}; C_n$	the Catalan generating function; Catalan numbers; a constant in (5.12); the central event.
$E_{m,n}(p); \mathbb{E}; \mathcal{E}_z$	the boundary error (4.3); expectation; the moment functional (5.8).
$H(\lambda); H_j; H_{20}$	the hook expression of Appendix A; Hermite polynomials; the constant (6.11).
β_j	recurrence coefficients (Section 6.2); not the exponent $\beta = -\alpha_m$ of the transseries volume.

2 Main results

Set

$$d = \binom{m}{2}, \quad J_m = \prod_{j=0}^{m-1} j!, \quad M_m(n) = \frac{(mn)!}{(n!)^m}, \quad \alpha_m = \frac{m^2 - 1}{2}, \quad (2.1)$$

and define

$$K_m = \frac{\sqrt{m} J_m}{4^d (2\pi)^{(m-1)/2}}. \quad (2.2)$$

The multinomial coefficient $M_m(n)$ counts all row-increasing arrays, before imposing any comparisons between different rows.

Theorem 2.1 (All orders at fixed height). *For every fixed integer $m \geq 1$ there is a uniquely determined sequence $c_{m,j} \in \mathbb{Q}$, with $c_{m,0} = 1$, such that, for every fixed $L \geq 0$,*

$$T_m(n) = M_m(n) \frac{J_m}{(4n)^d} \left(\sum_{j=0}^L \frac{c_{m,j}}{n^j} + O_{m,L}(n^{-L-1}) \right). \quad (2.3)$$

Every coefficient is given by the finite moment formula (5.9). In particular,

$$c_{m,1} = \frac{m(m^2 - 1)}{12}, \quad c_{m,2} = \frac{m^2(m^2 - 1)(m^2 + 2)}{288}. \quad (2.4)$$

Equivalently, there are rational $b_{m,j}$, $b_{m,0} = 1$, such that

$$T_m(n) = K_m m^{mn} n^{-\alpha_m} \left(\sum_{j=0}^L \frac{b_{m,j}}{n^j} + O_{m,L}(n^{-L-1}) \right), \quad (2.5)$$

where

$$b_{m,1} = \frac{(m^2 - 1)^2}{12m}, \quad (2.6)$$

$$b_{m,2} = \frac{(m^2 - 1)(m^6 + 3m^2 - 1)}{288m^2}. \quad (2.7)$$

Corollary 2.2 (A181199, with five corrections). *With $a(n) = T_5(n)$,*

$$a(n) = \frac{9 \cdot 5^{n+1/2}}{2^{17} \pi^2 n^{12}} \left(1 + \frac{48}{5n} + \frac{5233}{100n^2} + \frac{1028391}{5000n^3} + \frac{60248377}{100000n^4} + \frac{273939939}{250000n^5} + O(n^{-6}) \right). \quad (2.8)$$

Theorem 2.3 (Algebraicity dichotomy). *The ordinary generating function*

$$F_m(z) = \sum_{n \geq 0} T_m(n) z^n$$

is algebraic over $\mathbb{Q}(z)$ if and only if $m = 1$ or $m = 2$. The same classification holds for algebraicity over $\mathbb{C}(z)$.

Corollary 2.4 (A universal comparison with ordinary rectangles). *Let $f^{(n^m)}$ count ordinary standard Young tableaux of rectangular shape with m rows of length n . For fixed m ,*

$$\frac{T_m(n)}{f^{(n^m)}} = 2^{-m(m-1)} \left(1 + \frac{m(m^2 - 1)}{4n} + \frac{m^2(m^2 - 1)(m^2 - 2)}{32n^2} + O_m(n^{-3}) \right). \quad (2.9)$$

Thus the extra antidiagonal comparisons retain a positive limiting fraction $2^{-m(m-1)}$ of ordinary rectangular tableaux.

The constant is not obtained by treating the comparisons as independent. It comes from two shifted-hook denominators, one for each side of a threshold cut. Their product is the essential feature of the proof. The distributional counterpart is given in Section 9.

[write] Added 3 October 2026, batch 85: the comparison constant. The normalizer of Corollary 2.4 is the classical hook count of the rectangle, recorded in ProveIt as equation (3.19), label `t2:eq:tableaux`, of the volume *Combinatorial Transseries and Their Inverses* (Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/, Section 3.5), which writes it $T_d(n)$ with d rows; the hook product displayed at the end of Section 6.4 here is the same formula with $d = m$. That volume's constants (3.20) (`t2:eq:tableaux-constants`) give $f^{(n^m)} \sim C m^{mn} n^\beta$ with $a = m \log m$, $\beta = -(m^2 - 1)/2 = -\alpha_m$ and $C = \sqrt{m} (2\pi)^{(1-m)/2} \prod_{j=0}^{m-1} j!$. Comparing with (2.2), $K_m = 2^{-m(m-1)} C$ exactly, since $4^d = 2^{m(m-1)}$: the two families share the rate m^m and the exponent $-\alpha_m$, and the leading term of (2.9) is the quotient of the two constants. The intake checked this identity symbolically for $m = 2, \dots, 7$, and checked that (2.4) and the rising-factorial expansion of $\prod_{j < m} (n + j)!/n!$ give the two printed corrections of (2.9) for the same m . Neither the formula nor the constant of the ordinary rectangle is new here; the new statement is the shifted count and its ratio.

3 Prefix states and the shifted hook product

3.1 The exact state space

Fill an array by inserting $1, 2, \dots, mn$ in order. At any intermediate stage let x_i be the number of filled cells in row i . Row increase means that the filled cells in each row are a prefix. The legal states satisfy

$$n \geq x_1 \geq x_2 \geq \dots \geq x_m \geq 0, \quad x_i = x_{i+1} \implies x_i \in \{0, n\}. \quad (3.1)$$

Indeed, placing a filled cell in row $i + 1$, column $j < n$, forces the cell in row i , column $j + 1$, to have already been filled. When the lower row has length n , column increase forces the upper row to be full as well. Conversely, these conditions make the filled set an order ideal of the array poset. They therefore characterize the prefix states exactly.

A transition increases x_i by one. For $i > 1$ it is allowed precisely when

$$x_{i-1} \geq x_i + 2 \quad \text{or} \quad (x_i = n - 1 \text{ and } x_{i-1} = n), \quad (3.2)$$

with $x_i < n$; the first row has only the latter upper bound. Empty rows can tie, and completed rows can tie. Forgetting either exception would change the sequence.

For a legal state $\lambda = (\lambda_1, \dots, \lambda_m)$ define its rotated complement by

$$\lambda^c = (n - \lambda_m, \dots, n - \lambda_1). \quad (3.3)$$

Rotation through 180 degrees, together with reversal of the numerical labels, preserves the comparisons. Thus suffix enumeration at λ is prefix enumeration at λ^c .

3.2 Interior states

Call a state *interior* when

$$n > \lambda_1 > \dots > \lambda_m > 0. \quad (3.4)$$

Every path ending there has avoided the completed-row boundary. After removing trailing zero parts, it is a chain of strict partitions obtained by adding one box at a time. Such chains are standard shifted tableaux. If g^λ denotes their number and $k = |\lambda|$, the shifted hook product is

$$g^\lambda = \frac{k!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \quad (3.5)$$

This standard formula appears, for example, in Sun's account [8]. Appendix A proves it directly from the strict-partition recursion; no shifted-tableau enumeration result is needed without proof.

For an interior state both λ and λ^c are strict positive partitions. Consequently, the prefix and suffix counts are simultaneously available in product form. This is why cutting *in the middle*, rather than tracking the first completed row, is effective for asymptotics.

4 An exact random-threshold reduction

4.1 Order statistics and the cut

Take m independent rows, each consisting of n independent uniform random numbers in $(0, 1)$, sorted increasingly. Let $\mathcal{A}$ be the event that the comparisons between rows also hold. Every interleaving of these

sorted rows has the same probability, so

$$\mathbb{P}(\mathcal{A}) = \frac{T_m(n)}{M_m(n)}. \quad (4.1)$$

This is the order-statistics form of the order-polytope interpretation [8]; it also follows immediately by sorting all mn samples at once.

Fix $p \in (0, 1)$ and let X_i count the entries in row i below p . Before conditioning on $\mathcal{A}$, the variables X_i are independent $\text{Bin}(n, p)$. Conditional on these counts, the samples below p and above p are independent sorted uniforms in their respective intervals. When the count vector is an interior state λ , the conditional probability of satisfying all comparisons below the cut is

$$\frac{g^\lambda}{k! / \prod_i \lambda_i!}.$$

The analogous suffix probability uses λ^c . All comparisons across the cut are automatic because the prefix is an order ideal. Multiplying and using (3.5) gives

$$\mathbb{P}(\mathcal{A} \mid X = \lambda) = R_n(\lambda), \quad R_n(x) = \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)}. \quad (4.2)$$

The weight is symmetric in the coordinates. Equal interior coordinates give zero, and all distinct coordinates have exactly one decreasing order.

Theorem 4.1 (Interior sum and explicit boundary error). *Let $X_1, \dots, X_m$ be independent $\text{Bin}(n, p)$ variables and $\mathcal{I} = \{1 \leq X_i \leq n - 1 \text{ for all } i\}$. Then*

$$\frac{T_m(n)}{M_m(n)} = \frac{1}{m!} \mathbb{E} [\mathbf{1}_{\mathcal{I}} R_n(X)] + E_{m,n}(p), \quad (4.3)$$

where the error is nonnegative and satisfies

$$0 \leq E_{m,n}(p) \leq m(p^n + (1 - p)^n). \quad (4.4)$$

For $p = 1/2$ the expectation is the finite rational sum

$$I_m(n) = 2^{-mn} \sum_{n > x_1 > \dots > x_m > 0} \left(\prod_{i=1}^m \binom{n}{x_i} \right) \prod_{i < j} \frac{(x_i - x_j)^2}{(x_i + x_j)(2n - x_i - x_j)}, \quad (4.5)$$

and

$$0 \leq T_m(n) - M_m(n) I_m(n) \leq 2m 2^{-n} M_m(n). \quad (4.6)$$

Proof. The contribution to $\mathbb{P}(\mathcal{A})$ from interior count vectors is the sum of their binomial probabilities times (4.2). Symmetry converts this decreasing-vector sum into $1/m!$ times the unrestricted expectation on $\mathcal{I}$. The remaining event is $\mathcal{A} \cap \mathcal{I}^c$. It has nonnegative probability, at most the probability that some row has all its samples on one side of the cut. A union bound gives (4.4). The specialization is immediate. $\square$

Remark 4.2 (A useful pointwise bound). For $0 \leq x, y \leq n$, $|x - y| \leq x + y$ and $|x - y| \leq 2n - x - y$. Thus every factor of R_n lies in $[0, 1]$ whenever its denominator is nonzero. In particular, $0 \leq R_n \leq 1$ on $\mathcal{I}$. No uncontrolled singularity is hidden near the endpoints in (4.3).

The boundary estimate is an exact, nonasymptotic statement. It is deliberately simple and is not claimed sharp. Since the main probability is of order n^{-d} , this exponentially small additive error is negligible relative to every term of the inverse-power expansion.

5 Gaussian reduction and an effective all-order theorem

5.1 The scaled integrand

Use $p = 1/2$ and put

$$Y_i = \frac{2X_i - n}{\sqrt{n}}, \quad \Delta(y) = \prod_{i < j} (y_i - y_j), \quad q_{ij}(y) = (y_i + y_j)^2. \quad (5.1)$$

The pair weight becomes exactly

$$R_n(X) = (4n)^{-d} \Delta(Y)^2 \prod_{i < j} \left(1 - \frac{q_{ij}(Y)}{4n}\right)^{-1} \quad \text{on } \mathcal{I}. \quad (5.2)$$

Let $h_r(q)$ be the complete homogeneous polynomial of degree r in the d variables q_{ij} , defined by

$$\prod_{i < j} (1 - u q_{ij})^{-1} = \sum_{r \geq 0} h_r(q) u^r. \quad (5.3)$$

In particular $h_0 = 1$. For $m = 1$, the empty product is one and $h_r = 0$ for $r > 0$.

5.2 Polynomial standardized moments

A variable Y_i has the same law as $n^{-1/2}(\varepsilon_1 + \cdots + \varepsilon_n)$, where the ε_j are independent symmetric signs. Its moment generating function is

$$\mathbb{E} e^{tY_i} = (\cosh(t/\sqrt{n}))^n. \quad (5.4)$$

For every fixed nonnegative integer k , its k th moment is exactly a polynomial $\mu_k(z)$ at $z = 1/n$. One convenient formal definition is

$$\sum_{k \geq 0} \mu_k(z) \frac{t^k}{k!} = \exp \left(\frac{1}{z} \log \cosh(t\sqrt{z}) \right). \quad (5.5)$$

The apparent singularity at $z = 0$ is removable in the formal expansion. Odd moments vanish, and

$$\mu_0 = 1, \quad \mu_2 = 1, \quad \mu_4 = 3 - 2z, \quad (5.6)$$

$$\mu_6 = 15 - 30z + 16z^2, \quad \mu_8 = 105 - 420z + 588z^2 - 272z^3. \quad (5.7)$$

For $k = 2s > 0$, the degree is at most $s - 1$. This follows either by grouping terms according to equal sign indices, or from the moment–cumulant formula: a partition into even blocks contributes $z^{s-\#\text{blocks}}$ times a product of sign cumulants.

Define a linear functional $\mathcal{E}_z : \mathbb{Q}[y_1, \dots, y_m] \rightarrow \mathbb{Q}[z]$ by

$$\mathcal{E}_z(y_1^{k_1} \cdots y_m^{k_m}) = \prod_{i=1}^m \mu_{k_i}(z). \quad (5.8)$$

Then $\mathcal{E}_{1/n}$ is the exact expectation for the independent variables Y_i . At $z = 0$, it is expectation under independent standard normal variables.

5.3 The coefficient formula

Proposition 5.1 (Finite formula for each coefficient). *For $j \geq 0$, the coefficient in (2.3) is*

$$c_{m,j} = \frac{1}{m!J_m} \sum_{r=0}^j \frac{1}{4^r} [z^{j-r}] \mathcal{E}_z \left(\Delta(y)^2 h_r(q(y)) \right). \quad (5.9)$$

The formula uses finite polynomials and rational arithmetic. It does not require previously computed sequence terms or a guessed recurrence.

5.4 Uniform error control

We supply the estimates that justify the formal calculation. Let

$$C_n = \{|Y_i| \leq \sqrt{n}/2 \text{ for all } i\}.$$

Since $\cosh t \leq e^{t^2/2}$, Chernoff's inequality gives

$$\mathbb{P}(C_n^c) \leq 2me^{-n/8}. \quad (5.10)$$

The same moment generating function bound implies that the absolute moments of every fixed degree of the Y_i are bounded uniformly in n .

On C_n , each $q_{ij}(Y)/(4n)$ lies in $[0, 1/4]$. Taylor's theorem applied to the finite product in (5.2) therefore gives, for every fixed L ,

$$\prod_{i < j} \left(1 - \frac{q_{ij}(Y)}{4n} \right)^{-1} = \sum_{r=0}^L \frac{h_r(q(Y))}{(4n)^r} + \mathcal{R}_{L,n}(Y), \quad (5.11)$$

$$|\mathcal{R}_{L,n}(Y)| \leq C_{m,L} n^{-L-1} (1 + \|Y\|)^{2L+2}. \quad (5.12)$$

To see the second bound, differentiate the product $L + 1$ times with respect to $1/n$. Every differentiated term has total numerator degree $2L + 2$; all denominators are bounded below by fixed powers of $3/4$. After multiplication by $\Delta(Y)^2$, the expected remainder is still $O_{m,L}(n^{-L-1})$ by the uniform moment bounds.

We may replace polynomial expectations on C_n by unrestricted ones with exponentially small error. For example, Cauchy–Schwarz, (5.10), and a uniform moment bound control the difference for every fixed polynomial. For the original rational weight, use $R_n \leq 1$ from Remark 4.2. After restoring the factor $(4n)^d$, its discarded tail is only a fixed polynomial in n times an exponential. The endpoint error (4.4) behaves similarly. We have consequently proved

$$\frac{T_m(n)}{M_m(n)} = \frac{(4n)^{-d}}{m!} \left(\sum_{r=0}^L \frac{\mathcal{E}_{1/n}(\Delta^2 h_r)}{(4n)^r} + O_{m,L}(n^{-L-1}) \right). \quad (5.13)$$

This establishes (5.9) after collecting powers. It also proves that parity effects cannot occur in the algebraic sector: the moments are exact polynomials in $1/n$, on both parity subsequences.

5.5 The leading Gaussian integral

For independent standard normals $G_1, \dots, G_m$,

$$\mathbb{E} \Delta(G)^2 = m! \prod_{j=0}^{m-1} j! = m! J_m. \quad (5.14)$$

Here is an elementary verification. Replace the monomials in the Vandermonde determinant by the monic Hermite polynomials H_j , whose squared norms for standard normal measure are $j!$. The determinant is unchanged because this basis change is unit triangular. Expanding the square and using orthogonality leaves the $m!$ matching permutations, each contributing $\prod j!$.

Thus $c_{m,0} = 1$. Finally, Stirling's expansion gives

$$M_m(n) \sim \sqrt{m} (2\pi n)^{(1-m)/2} m^{mn}, \quad (5.15)$$

which proves the leading constant and exponent in (2.5). More explicitly, the complete conversion between the two normalizations is

$$\sum_{j \geq 0} b_{m,j} z^j = \left(\sum_{j \geq 0} c_{m,j} z^j \right) \exp \left(\sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)} (m^{1-2r} - m) z^{2r-1} \right), \quad (5.16)$$

where B_{2r} are Bernoulli numbers. All uses of this identity are formal, or truncated to a fixed finite order with Stirling's remainder estimate [11]. This completes the proof of Theorem 2.1, apart from the explicit evaluations in the next section.

6 Universal first and second corrections

The finite formula is already an all-order algorithm. To obtain short formulas valid at every height, it is better to keep the Vandermonde weight intact rather than expand it into individual monomials.

6.1 An exact binomial Vandermonde normalization

Define the normalized binomial ensemble by weighting the independent Y_i with $\Delta(Y)^2$. Its normalizing factor is

$$Z_m(n) = \mathbb{E} \Delta(Y)^2 = m! J_m \prod_{j=0}^{m-1} \frac{n^j}{n^j}. \quad (6.1)$$

This identity is valid for $n \geq m-1$ (and interpreted as zero when the support cannot accommodate m distinct points).

For completeness, let $X \sim \text{Bin}(n, 1/2)$ and define $K_j(X)$ by

$$(1+t)^X (1-t)^{n-X} = \sum_{j=0}^n K_j(X) t^j.$$

Multiplying two such generating functions and taking expectation gives $(1+st)^n$. Hence the K_j are orthogonal and $\mathbb{E} K_j^2 = \binom{n}{j}$. Their leading coefficient as polynomials in $Y = (2X - n)/\sqrt{n}$ is $n^{j/2}/j!$. The corresponding monic polynomials therefore have squared norms

$$h_j = j! n^{j/2}/j!.$$

The same determinant argument as in (5.14) proves (6.1). These are the symmetric Krawtchouk polynomials; the standard orthogonal-polynomial framework is recorded in [12].

Put

$$A = \binom{m}{3}, \quad B = \frac{m(m-1)^2(m-2)}{12}. \quad (6.2)$$

Expanding the finite product in (6.1) gives

$$\frac{Z_m(n)}{m!J_m} = 1 - \frac{A}{n} + \frac{A^2 - B}{2n^2} + O_m(n^{-3}). \quad (6.3)$$

Indeed, $A = \sum_{j < m} \sum_{k < j} k$ and $B = \sum_{j < m} \sum_{k < j} k^2$.

6.2 The first pair statistic

Let $p_r = \sum_i Y_i^r$. The first complete homogeneous polynomial is

$$S_1 = h_1(q) = \sum_{i < j} (Y_i + Y_j)^2 = (m-2)p_2 + p_1^2. \quad (6.4)$$

Write $\langle \cdot \rangle_n$ for expectation in the normalized binomial Vandermonde ensemble. The monic polynomial recurrence is

$$yP_j(y) = P_{j+1}(y) + \beta_j P_{j-1}(y), \quad \beta_j = j \left(1 - \frac{j-1}{n}\right).$$

The diagonal coefficient is zero by symmetry. The norm quotient proves the displayed value of β_j .

We need the standard finite-projection identities

$$\langle p_2 \rangle_n = \beta_m + 2 \sum_{j=1}^{m-1} \beta_j = m^2 - \frac{m(m-1)(2m-1)}{3n}, \quad (6.5)$$

$$\langle p_1^2 \rangle_n = \beta_m = m - \frac{m(m-1)}{n}. \quad (6.6)$$

They can be seen directly from the determinant ensemble: multiplication by y is tridiagonal in the orthonormal basis, with off-diagonal entries $\sqrt{\beta_j}$. The mean of a linear statistic is its multiplication operator's trace on degrees $0, \dots, m-1$. Its variance is the squared norm of the part crossing that projection. For p_1 only the edge from degree $m-1$ to degree m crosses, giving β_m . These trace rules also follow by expanding $\mathbb{E} \prod_i (1 + tf(Y_i))$ as a determinant of the moment matrix and taking the first two derivatives at $t = 0$.

Consequently,

$$\langle S_1 \rangle_n = s_{10} + \frac{s_{11}}{n}, \quad s_{10} = m(m-1)^2, \quad s_{11} = -\frac{m(m-1)(2m^2 - 5m + 5)}{3}. \quad (6.7)$$

6.3 The second pair statistic in the Gaussian limit

Let $\langle \cdot \rangle_0$ denote the Gaussian Vandermonde ensemble. Since

$$h_2(q) = \frac{1}{2} \left(S_1^2 + \sum_{i < j} (Y_i + Y_j)^4 \right),$$

and

$$\sum_{i < j} (Y_i + Y_j)^4 = (m-8)p_4 + 4p_1p_3 + 3p_2^2,$$

we need only a few Gaussian moments:

$$\langle p_2^2 \rangle_0 = m^2(m^2 + 2), \quad \langle p_2p_1^2 \rangle_0 = m(m^2 + 2), \quad (6.8)$$

$$\langle p_1^4 \rangle_0 = 3m^2, \quad \langle p_4 \rangle_0 = 2m^3 + m, \quad \langle p_1p_3 \rangle_0 = 3m^2. \quad (6.9)$$

There is no need to invoke random-matrix integration to establish these. Scaling and translation of the Gaussian integral give

$$\langle e^{tp_2+sp_1} \rangle_0 = (1-2t)^{-m^2/2} \exp\left(\frac{ms^2}{2(1-2t)}\right). \quad (6.10)$$

This proves the first three identities by differentiation. Translation integration by parts, using $\Delta(y+c\mathbf{1}) = \Delta(y)$, gives $\langle p_1 p_3 \rangle_0 = 3\langle p_2 \rangle_0 = 3m^2$. Finally, in the Hermite multiplication matrix, the degree- j diagonal entry of the fourth power is $6j^2 + 6j + 3$; summing over $0 \leq j < m$ yields $2m^3 + m$. Thus

$$H_{20} := \langle h_2(q) \rangle_0 = \frac{m(m-1)(m^4 - 3m^3 + 10m^2 - 18m + 16)}{2}. \quad (6.11)$$

6.4 Collecting the terms

Combining (6.3), (6.7), and (6.11),

$$c_{m,1} = -A + \frac{s_{10}}{4}, \quad (6.12)$$

$$c_{m,2} = \frac{A^2 - B}{2} + \frac{-As_{10} + s_{11}}{4} + \frac{H_{20}}{16}. \quad (6.13)$$

Straight polynomial simplification gives (2.4). The first Stirling logarithm coefficient is $u_1 = (m^{-1} - m)/12$, so $b_{m,1} = c_{m,1} + u_1$ and $b_{m,2} = c_{m,2} + u_1 c_{m,1} + u_1^2/2$. This proves (2.7).

The ordinary rectangular hook product is

$$f^{(n^m)} = M_m(n) J_m \prod_{j=0}^{m-1} \frac{n!}{(n+j)!}.$$

Its finite rising-factorial expansion, combined with (2.4), proves (2.9).

7 OEIS specializations and numerical checks

7.1 Low-height controls

For $m = 1$, $T_1(n) = 1$. For $m = 2$, the usual strict-ballot argument gives $T_2(n) = C_{n-1}$ for $n \geq 1$; equivalently,

$$T_2(n) = \binom{2n}{n} \frac{1}{2(2n-1)} = \binom{2n}{n} \frac{1}{4n} \sum_{j \geq 0} \frac{1}{(2n)^j}. \quad (7.1)$$

Thus $c_{2,j} = 2^{-j}$ at every order. This is a useful exact control of both the normalization and the coefficient generator.

For height three, the theorem recovers the existing A181197 leading term [2] and gives

$$T_3(n) = \frac{\sqrt{3} 27^n}{64\pi n^4} \left(1 + \frac{16}{9n} + \frac{755}{324n^2} + O(n^{-3}) \right). \quad (7.2)$$

For height four, corresponding to A181198,

$$T_4(n) = \frac{3 \cdot 256^n}{1024\sqrt{2} \pi^{3/2} n^{15/2}} \left(1 + \frac{75}{16n} + \frac{6905}{512n^2} + O(n^{-3}) \right). \quad (7.3)$$

The leading constants are consequences of the general theorem, not inputs fitted from these sequences.

7.2 Five corrections in multinomial normalization

Table 1 reports exact output of (5.9). Only the compact multinomial normalization is used in this table. The archive also supplies the corresponding $b_{m,j}$.

Table 1: Coefficients $c_{m,j}$ in (2.3); $c_{m,0} = 1$.

m	$c_{m,1}$	$c_{m,2}$	$c_{m,3}$	$c_{m,4}$	$c_{m,5}$
2	1/2	1/4	1/8	1/16	1/32
3	2	11/4	23/8	45/32	-57/16
4	5	15	33	789/16	93/16
5	10	225/4	1819/8	22045/32	1353
6	35/2	665/4	8939/8	92351/16	731621/32

7.3 How much the corrections improve A181199

For $m = 5$, let $L_5(n) = K_5 5^{5n} n^{-12}$ and $A_{5,L}(n) = L_5(n) \sum_{j=0}^L b_{5,j} n^{-j}$. Table 2 shows the signed relative error $A_{5,L}(n)/T_5(n) - 1$. Its values are numerical diagnostics, not rigorous interval enclosures. The counts at $n = 10, 20, 40$ were independently regenerated by the row-state dynamic program and compared with OEIS; $n = 80$ is an OEIS reference value, not regenerated in this run [6].

Table 2: A181199: signed relative errors of successive truncations.

n	$L = 0$	$L = 1$	$L = 2$	$L = 5$
10	$-6.37190 \cdot 10^{-1}$	$-2.88893 \cdot 10^{-1}$	$-9.90345 \cdot 10^{-2}$	$1.42184 \cdot 10^{-3}$
20	$-3.90471 \cdot 10^{-1}$	$-9.78967 \cdot 10^{-2}$	$-1.81550 \cdot 10^{-2}$	$1.97547 \cdot 10^{-5}$
40	$-2.16403 \cdot 10^{-1}$	$-2.83393 \cdot 10^{-2}$	$-2.71080 \cdot 10^{-3}$	$2.67266 \cdot 10^{-7}$
80	$-1.13941 \cdot 10^{-1}$	$-7.61418 \cdot 10^{-3}$	$-3.69269 \cdot 10^{-4}$	$3.76974 \cdot 10^{-9}$

The leading term alone converges rather slowly at height five because $b_{5,1} = 9.6$. This explains why the constant is difficult to recognize reliably from modestly sized data without correction terms. The proof, however, does not use the numerical agreement.

8 A complete algebraicity classification

We now prove Theorem 2.3. The argument uses the classical Newton–Puiseux property: an algebraic function has a convergent fractional power expansion at a finite point, with algebraic coefficients when both the defining polynomial and the expansion point are algebraic. It has no logarithmic leading singularity. This is standard algebraic-function background; see the algebraic-function treatment in [10].

Put $R_m = m^{-m}$ and rescale

$$G_m(z) = F_m(R_m z) = \sum_{n \geq 0} v_n z^n, \quad v_n = R_m^n T_m(n).$$

For $m \geq 2$, Theorem 2.1 gives

$$v_n = K_m n^{-\alpha_m} (1 + O(n^{-1})). \quad (8.1)$$

In particular the radius of convergence of G_m is exactly one.

8.1 Odd heights: a logarithmic obstruction

Suppose $m \geq 3$ is odd. Then α_m is a positive integer. Set $q = \alpha_m - 1$. The coefficient of z^k in $G_m^{(q)}(z)$ is

$$\frac{(k+q)!}{k!} v_{k+q} = \frac{K_m}{k} + O(k^{-2}) \quad (k \rightarrow \infty).$$

After a harmless adjustment of finitely many terms, subtraction of $K_m \sum_{k \geq 1} z^k/k$ leaves an absolutely summable coefficient sequence. Therefore, along the real interval $z \uparrow 1$,

$$G_m^{(q)}(z) = K_m \log \frac{1}{1-z} + O(1). \quad (8.2)$$

A nonzero logarithmic divergence is incompatible with a Puiseux expansion. If F_m were algebraic, so would be its rescaling and every derivative; (8.2) is a contradiction.

8.2 Even heights: a transcendental-amplitude obstruction

Suppose $m \geq 4$ is even. Now $q = \alpha_m - 1/2 = (m^2 - 2)/2$ is an integer. The derivative coefficients satisfy

$$[z^k] G_m^{(q)}(z) = K_m k^{-1/2} + O(k^{-3/2}).$$

Since $[z^k](1-z)^{-1/2} = 1/\sqrt{\pi k} + O(k^{-3/2})$, coefficient subtraction again leaves an absolutely summable series. Hence

$$G_m^{(q)}(z) = K_m \sqrt{\pi} (1-z)^{-1/2} + O(1), \quad z \uparrow 1. \quad (8.3)$$

If the function were algebraic over $\mathbb{Q}(z)$, the leading coefficient in this Puiseux expansion at the algebraic point $z = 1$ would be algebraic. But

$$K_m \sqrt{\pi} = \frac{\sqrt{m} J_m}{4^d 2^{(m-1)/2}} \pi^{-(m-2)/2} \quad (8.4)$$

is a nonzero algebraic multiple of a nonzero integer power of π^{-1} . It is transcendental: otherwise that power of π , and hence π itself, would be algebraic. This contradiction uses only the classical transcendence of π .

8.3 The two algebraic cases and descent of constants

The exceptional cases are explicit:

$$F_1(z) = \frac{1}{1-z}, \quad F_2(z) = 1 + zC(z), \quad C(z) = \frac{1 - \sqrt{1-4z}}{2z}.$$

Thus $m = 1, 2$ are algebraic.

Finally, a rational-coefficient power series algebraic over $\mathbb{C}(z)$ is algebraic over $\mathbb{Q}(z)$. Indeed, fix finite degree bounds for a nonzero polynomial relation. Equating all coefficients gives a homogeneous linear system with rational entries in finitely many unknown polynomial coefficients. The descending chain of solution spaces stabilizes after finitely many independent equations. A nonzero complex solution therefore implies a nonzero rational solution. This proves the asserted classification over either field.

Remark 8.1. Nonalgebraicity does not imply non-D-finiteness. Logarithmic singularities, and half-integer singularities with transcendental connection constants, are compatible with D-finite functions. The guessed recurrences for heights four and five remain separate questions.

9 Gaussian laws for intermediate shapes

The same reduction describes typical partially filled arrays, not only their total number. We give both a randomized numerical cut and a fixed number of inserted labels; these are different probability models.

9.1 A cut at a fixed numerical level

Choose a tableau uniformly among the $T_m(n)$ arrays. Independently choose mn uniform random numbers in $(0, 1)$, sort them, and place them in the array in increasing order of its integer labels. This produces the uniform order-statistics model conditioned on $\mathcal{A}$. For a fixed $p \in (0, 1)$ let X_i be the number of row- i entries below p and set

$$Z_i = \frac{X_i - np}{\sqrt{np(1-p)}}. \quad (9.1)$$

Theorem 9.1 (Gaussian Vandermonde limit). *For fixed m and fixed $p \in (0, 1)$, the decreasing vector Z converges in distribution to the probability density*

$$\frac{e^{-\|z\|^2/2} \Delta(z)^2}{(2\pi)^{m/2} J_m}, \quad z_1 > z_2 > \cdots > z_m, \quad (9.2)$$

with respect to m -dimensional Lebesgue measure. Polynomial moments of fixed degree converge as well.

Proof. For interior decreasing x , the exact conditional mass is

$$\frac{\prod_i \mathbb{P}(\text{Bin}(n, p) = x_i) R_n(x)}{T_m(n)/M_m(n)}.$$

In bounded sets of standardized coordinates, the local central limit estimate for the binomial mass follows directly from Stirling's formula. Moreover,

$$x_i - x_j = \sqrt{np(1-p)}(z_i - z_j),$$

whereas the two denominator factors in (4.2) are asymptotic to $2np$ and $2n(1-p)$. Thus

$$R_n(x) = (4n)^{-d} \Delta(z)^2 (1 + o(1))$$

uniformly on every bounded set. Divide by $T_m(n)/M_m(n) \sim J_m(4n)^{-d}$ and use the lattice mesh $1/\sqrt{np(1-p)}$ in every coordinate. The Riemann sums converge to (9.2).

To justify passage beyond bounded sets, first restrict every x_i/n to a fixed compact interval inside $(0, 1)$ containing p . There the pair denominators are bounded below by a constant times n^2 , and binomial local bounds give a Gaussian upper bound times a fixed polynomial in the standardized coordinates. This is integrable, with every polynomial moment. Outside that macroscopic interval, Chernoff bounds and $R_n \leq 1$ give an exponentially small contribution even after division by the order- n^{-d} conditioning probability. The boundary term satisfies the same conclusion by (4.4). This proves tightness and moment convergence. Normalization of the limit is (5.14), with the factor $m!$ removed on the decreasing chamber. $\square$

The density (9.2) is the ordered eigenvalue density of the Gaussian unitary ensemble in the normalization with Gaussian factor $e^{-\sum z_i^2/2}$. This terminology is optional: the theorem and its normalization have already been specified by an explicit density.

9.2 A deterministic fraction of the labels

Let $\Lambda_i(k)$ be the number of cells in row i occupied by labels $1, \dots, k$ of a uniformly random tableau. Fix $p \in (0, 1)$ and take $k = \lfloor mnp \rfloor$.

Theorem 9.2 (Trace-zero limit at deterministic time). *The vector*

$$\left(\frac{\Lambda_i(k) - np}{\sqrt{np(1-p)}} \right)_{i=1}^m$$

converges to the density

$$\frac{e^{-\|z\|^2/2} \Delta(z)^2}{(2\pi)^{(m-1)/2} J_m} \quad (9.3)$$

on $\{z_1 > \dots > z_m, \sum_i z_i = 0\}$, with respect to Euclidean $(m-1)$ -dimensional measure on the trace-zero hyperplane.

Proof. In the preceding randomized-cut construction, the total number K of samples below p is $\text{Bin}(mn, p)$ and is independent of the integer tableau. Conditioning on $K = k$ therefore gives precisely $\Lambda(k)$. For an interior vector x of total size k , divide the mass used in the preceding proof by $\mathbb{P}(K = k)$. The latter is asymptotic to $(2\pi mnp(1-p))^{-1/2}$. The lattice $\{x \in \mathbb{Z}^m : \sum x_i = k\}$ has Euclidean covolume $\sqrt{m}$. In standardized coordinates its cell volume is $\sqrt{m} (np(1-p))^{-(m-1)/2}$. These factors give exactly the density (9.3) under the local binomial estimate.

For domination, use the same compact macroscopic region as before. The product of local binomial upper bounds on the hyperplane, after the additional conditioning, is bounded by a fixed polynomial times a Gaussian in the standardized coordinates, times the hyperplane lattice-cell volume. Its tails are uniformly summable. Outside the macroscopic region, the extra factor $1/\mathbb{P}(K = k) = O(\sqrt{n})$ cannot overcome an exponential Chernoff bound. This proves convergence of the hyperplane Riemann sums.

Finally, Δ is invariant under common translation. Orthogonally splitting $\mathbb{R}^m$ into the trace-zero hyperplane and the unit trace direction shows that the full Gaussian Vandermonde integral is $\sqrt{2\pi}$ times the trace-zero integral. Equation (5.14) then verifies the normalization in (9.3). $\square$

This theorem concerns one fixed macroscopic time. It is not a claim of process convergence to a noncolliding Brownian bridge; proving tightness in a path space and identifying the joint time laws require additional work.

10 Ratios, eventual log-convexity, and inversion

10.1 Consecutive ratios

Write $\alpha = \alpha_m$ and $b_1 = b_{m,1}$. For $m \geq 2$, the all-order expansion gives

$$\frac{T_m(n+1)}{T_m(n)} = m^m \left(1 - \frac{\alpha}{n} + \frac{\alpha(\alpha+1)/2 - b_1}{n^2} + O_m(n^{-3}) \right). \quad (10.1)$$

Hence $T_m(n)$ is eventually strictly increasing. Expanding $\log T_m(n)$ at $n-1, n, n+1$ with enough terms also gives

$$\log T_m(n+1) - 2 \log T_m(n) + \log T_m(n-1) = \frac{\alpha}{n^2} + O_m(n^{-3}) > 0 \quad (10.2)$$

for sufficiently large n . Thus it is eventually strictly log-convex. These statements do not identify the smallest valid n .

10.2 Inverting the asymptotic profile

Put $\lambda = m \log m$, $K = K_m$, and $L = \log(y/K)$. The large solution of

$$\lambda x - \alpha \log x = L$$

is

$$x_0 = -\frac{\alpha}{\lambda} W_{-1} \left(-\frac{\lambda}{\alpha} e^{-L/\alpha} \right), \quad (10.3)$$

where W_{-1} is the real branch taking values at most -1 [13]. The branch choice matters: the principal branch describes the small, irrelevant solution. In elementary logarithmic form,

$$x_0 = \frac{L + \alpha \log L - \alpha \log \lambda}{\lambda} + O_m \left(\frac{\log L}{L} \right). \quad (10.4)$$

Let $\ell_2 = b_{m,2} - b_{m,1}^2/2$. Formal inversion of the continuous asymptotic profile gives

$$x = x_0 - \frac{b_{m,1}}{\lambda x_0} - \frac{\alpha b_{m,1}/\lambda^2 + \ell_2/\lambda}{x_0^2} + O_m(x_0^{-3}). \quad (10.5)$$

To verify it, insert $x = x_0 + \delta$ into $\lambda x - \alpha \log x + b_{m,1}/x + \ell_2/x^2 = L$ and compare the coefficients of x_0^{-1} and x_0^{-2} . Along the exact values $y = T_m(n)$, the approximation in (10.5) returns $n + O_m(n^{-3})$. The higher inverse terms are obtained by the same recursive substitution. This inversion technique is standard and also used elsewhere in ProveIt; its application here is a consequence of the new expansion, not a new inversion method.

For an arbitrary target y , the least integer index attaining it is a staircase function. An asymptotic estimate alone must not be rounded and advertised as an exact threshold certificate. Near an integer transition, explicit upper and lower bounds for the neighboring sequence values are required. The finite interior sum and its boundary error (4.6) supply one possible, though not always computationally efficient, route to such bounds.

[write] Added 3 October 2026, batch 85: the inversion is an instance. The inversion (10.3) is a case of Theorem J.23 (`p0:thm:lambert-core`, “Exact solution of the dominant block”) of the canonical transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`, which solves $aX + b \log X = L$ by $X = (b/a) W_k((a/b)e^{L/b})$. Here $a = \lambda = m \log m > 0$ and $b = -\alpha_m < 0$ for $m \geq 2$, and that theorem’s branch rule for $b < 0$ gives exactly (10.3): the large root is the W_{-1} root, and it exists precisely when $L > L_c = \alpha_m(1 - \log(\alpha_m/\lambda))$ (for $m = 5$, $K_5 e^{L_c} \approx 0.021$, so every $y \geq 1$ qualifies). The text’s own sentence that the technique “is standard and also used elsewhere in ProveIt” is accurate; nothing in this section is a new inversion result. The real substitution, branch rules, threshold and slope of Theorem J.23 are formalized in `Analysis/FabiusFunction/Lean/FabiusFunction/LinLogCoreInversion.lean` (the volume’s register rates the theorem “Partial”: the asymptotic and general complex-branch clauses are not formalized); that formalization concerns the general equation, and nothing specific to shifted rectangles is formalized. The caution about integer thresholds in the preceding paragraph is the separation condition of the volume’s staircase theorem, Theorem J.43 (`p0:thm:staircase`). The path-forest report cited in Section 1.3 inverts its sequences by the same apparatus (its `spf:thm:inverse-first` and `spf:thm:inverse-all`).

11 Reproducibility and computational scope

11.1 Exact coefficient generation

The program `code/coefficients.py` implements (5.9) using Python’s exact rational arithmetic. It avoids expanding the squared Vandermonde in full. For any symmetric polynomial P and any product measure with identical one-variable moments,

$$\mathcal{E}_z(\Delta^2 P) = m! \sum_{\sigma \in S_m} \text{sgn}(\sigma) \mathcal{E}_z \left(P \prod_{i=0}^{m-1} y_{i+1}^{i+\sigma(i)} \right). \quad (11.1)$$

Expand the two Vandermonde determinants, permute the integration variables in one expansion, and use symmetry of P to prove this identity. Only $m!$ permutations remain. Equal sorted exponent profiles are grouped before evaluating products of moment polynomials.

The h_r are generated from (5.3); the standardized moments are generated from sign cumulants. The Stirling conversion (5.16) is also done with rational numbers. The method is practical for modest fixed m and moderate order. Its permutation factor makes it unsuitable as presently implemented for large m ; no claim of polynomial complexity in m is made.

11.2 Independent enumeration and exact finite identities

The program `code/tableaux.py` supplies two enumerators: a row-state dynamic program using (3.2), and an independent bitmask linear-extension enumerator using the cell-level predecessor relation. The latter is intentionally restricted to small arrays. Neither uses a conjectured recurrence.

The verification run recorded in the archive passed 471 checks in the following suites:

Suite	Checks
Bitmask-poset versus row-state enumeration	27
Interior prefix counts versus the shifted hook product	238
Threshold split, rational interior sum, and boundary inequality	120
Exact Catalan control, $n = 1, \dots, 30$	30
Selected A181198 values, independently regenerated	12
Selected A181199 values, independently regenerated	12
Universal corrections versus the coefficient engine, $m = 1, \dots, 6$	30
Standardized binomial-moment controls	2

The independently regenerated OEIS comparisons use $n = 1, \dots, 10, 20, 40$ for each of heights four and five [5, 6]. The threshold identity is checked for $1 \leq m \leq 5$ and $1 \leq n \leq 8$. These are finite corroborations of the proofs, not substitutes for them.

11.3 Commands and data provenance

From the archive’s top-level directory:

```
python3 code/verify.py
python3 code/coefficients.py --height 5 --order 5
```

```
python3 code/numerics.py
bash build.sh
```

The first two commands use the Python standard library only. The numerical diagnostics additionally use `mpmath`; they were run with 90 decimal digits. The PDF build requires a conventional LaTeX installation with the packages listed in the preamble.

The reference integers in `data/oeis_selected.json` were transcribed from the entry pages and b-files inspected during the source audit. They are not outputs of the coefficient engine. Counts regenerated independently are saved separately in `data/exact_regenerated.json`. Values at $n = 80$ appear only as reference inputs to numerical diagnostics. Full third-party articles are not redistributed in this archive.

[write] Added 3 October 2026, batch 85: the shipped layout. “The archive” in this section is the delivered package. In Provelt its files keep the subdirectories `code/` and `data/`, except four, which moved as follows:

Delivered as	Shipped as
<code>build.sh</code>	<code>code/build.sh</code>
<code>requirements-numeric.txt</code>	<code>data/requirements-numeric.txt</code>
<code>notes/SOURCES_AND_STATUS.md</code>	<code>notes-SOURCES_AND_STATUS.md</code>
<code>notes/PROPOSED_OEIS_ADDITIONS.txt</code>	<code>data/notes-PROPOSED_OEIS_ADDITIONS.txt</code>

`code/build.sh` changes to its own directory, so in its shipped place it no longer finds `article.tex`. The delivered PDF and checksum list are not shipped; they survive in the arrival commit’s archive. `code/verify.py` and `code/numerics.py` write their outputs into the report’s own `data/` directory, overwriting the shipped records, so they must be rerun on a copy; the README of the report gives the commands. At intake (3 October 2026, Python 3.14.4 on Windows) `code/verify.py` passed all 471 checks on a copy, and every regenerated data file agreed with the shipped one after removal of carriage returns, except the timing field of `data/verification.json`; `code/coefficients.py` and `code/numerics.py` (`mpmath` 1.3.0) reproduced `data/coefficients_m5.json` and `data/numerics.csv`.

`data/oeis_selected.json` holds terms of A181198 and A181199 copied from the OEIS entries and b-files (Koutschan, with earlier terms by Heinz); OEIS content is published by The OEIS Foundation Inc. under the Creative Commons Attribution-ShareAlike 4.0 licence, so this file is third-party data under that licence, not MIT-0. The draft OEIS text in `data/notes-PROPOSED_OEIS_ADDITIONS.txt` is marked “DRAFT ONLY — NOT SUBMITTED” and stays so: nothing was submitted to OEIS by the package’s author or by the intake, and its own instruction not to mark either conjectured recurrence as proved stands.

12 Further research questions and proposed approaches

The following questions are not presented as solved. They are chosen to continue the present line of work rather than to restate the resolved leading-term conjecture.

Research question 1 (Prove the specific height-four and height-five recurrences). Can the exact enumeration be transformed into a finite hypergeometric sum or a Pfaffian identity that admits a compact creative-telelescoping certificate for the recurrences in A181198 and A181199?

The random-threshold expression is exact only after adding its boundary part. For a recurrence proof that part cannot be discarded. A useful first step is an exact decomposition according to the number of full and empty rows at the cut, followed by shifted-skew-tableau formulas for those boundary states. A proof should compare the resulting operator with the published guessed operator, not merely generate more confirming terms.

Research question 2 (Polynomial dependence of every correction on height). For each fixed j , is $c_{m,j}$ a polynomial in m of degree at most $3j$? Can its factorization and denominator structure be described uniformly?

The formulas for $j = 1, 2$ support this degree scale. The question beyond those orders is not proved here. A promising approach is to reorganize (5.9) as Gaussian matrix moments or as connected cumulant diagrams, in which the powers of m count index loops. This could replace factorial-in- m computation with symbolic fixed-order formulas.

Research question 3 (A growing-height transition). For what growth rates $m = m(n)$ does the leading fixed-height formula remain relatively accurate? Is the natural first transition governed by m^3/n ?

The coefficient $c_{m,1} \sim m^3/12$ suggests this parameter in multinomial normalization. It does not establish a uniform regime. A proof would need uniform Vandermonde moment bounds and a growing-dimension tail analysis. For m^3/n of order one, connected logarithmic corrections may require resummation rather than ordinary fixed-order truncation.

Research question 4 (Exponentially small sectors and sharp boundary asymptotics). What is the true leading asymptotic of the discarded boundary term $E_{m,n}(1/2)$, and how does it combine with large deviations of the interior expectation?

The bound $2m2^{-n}$ is sufficient for every inverse power but is not an asymptotic formula. Configurations with a row empty or full give a natural finite hierarchy of lower-dimensional boundary strata. Computing their constants could produce a genuine transseries beyond the algebraic sector, and could clarify the signs and optimal truncation of the coefficients.

Research question 5 (A full path-process limit). Do the centered row-length processes of a uniform tableau converge to a trace-zero noncolliding Gaussian bridge on compact time intervals inside $(0, 1)$? What are the entrance and exit scalings near the endpoints?

Theorem 9.2 establishes one-time marginals only. A proof of process convergence would require multi-cut enumeration, control of increments, and tightness. The complement symmetry provides a useful constraint on any proposed entrance/exit description.

Research question 6 (Certified asymptotic enclosures and integer thresholds). Can the moment expansion be packaged into efficient explicit upper and lower bounds with practical constants, avoiding the $\binom{n-1}{m}$ -term interior sum?

On a central region the pair interaction has a positive geometric expansion. Combining its finite partial sums with concentration bounds and exact moment determinants could give certified intervals. These would turn the inverse approximations into exact index decisions away from a quantified transition zone.

Research question 7 (D-finiteness versus nonalgebraicity at arbitrary height). Is $F_m(z)$ D-finite for every fixed m , and how does the minimal annihilating order grow with m ?

Theorem 2.3 settles the algebraic/nonalgebraic distinction, not the D-finite/non-D-finite one. An exact finite-dimensional differential system in the fixed-height regime would be a stronger structural result. The opposite-orientation constant-width transfer matrix does not answer this question.

13 Conclusion

A shifted rectangle admits a particularly effective asymptotic cut: both sides of an interior threshold state are ordinary strict shifted shapes. The two shifted hook products turn the count into a binomial Vandermonde expectation, with an explicitly controlled boundary error. This gives a proof of the A181199 leading constant independent of all guessed recurrences, an effective all-order expansion for every fixed height, and universal correction polynomials through second order.

The consequences go beyond adding coefficients to an OEIS entry. The same constant yields a complete algebraicity dichotomy for the fixed-height generating functions, while the same exact weight identifies Gaussian laws for typical intermediate shapes. The remaining recurrence, uniformity, and exponentially small questions are separate, concrete targets with clear starting points.

A A direct proof of the shifted hook product

Let $\lambda_1 > \dots > \lambda_r > 0$ be a strict partition of k , and put

$$H(\lambda) = \frac{k!}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}.$$

We prove that it satisfies the strict-partition path recursion

$$H(\lambda) = \sum_i H(\lambda - e_i), \quad (\text{A.1})$$

where only valid strict-partition removals are included and a trailing zero part is deleted. Together with $H(\emptyset) = 1$, this proves (3.5) by induction.

For a valid removal,

$$\frac{H(\lambda - e_i)}{H(\lambda)} = \frac{\lambda_i}{k} \prod_{j \neq i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1}. \quad (\text{A.2})$$

The same expression is zero for an invalid removal causing equal positive parts. If the last part is one, deleting its resulting zero gives the same ratio, since the extra zero factors cancel. Thus it suffices to prove

$$\sum_i \lambda_i \prod_{j \neq i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1} = k. \quad (\text{A.3})$$

Consider the rational function

$$R(z) = \prod_{j=1}^r \frac{z - \lambda_j - 1}{z - \lambda_j} \frac{z + \lambda_j}{z + \lambda_j - 1}, \quad f(z) = (z - \frac{1}{2})R(z).$$

Its possible finite poles are the disjoint sets λ_i and $1 - \lambda_i$. Direct substitution gives $R(1 - z) = R(z)$ and hence $f(1 - z) = -f(z)$. The residues at the paired poles are equal. At $z = \lambda_i$, the residue is the negative of the i th summand in (A.3).

At infinity, each pair of factors contributes $-2\lambda_j/z^2$ to the first nonconstant term, so

$$R(z) = 1 - \frac{2k}{z^2} + O(z^{-3}), \quad f(z) = z - \frac{1}{2} - \frac{2k}{z} + O(z^{-2}).$$

The sum of finite residues is the coefficient of z^{-1} , namely $-2k$. On the other hand it is minus twice the left side of (A.3). The identity follows, proving (A.1) and the hook product. All operations here are rational identities; the residue language can equivalently be replaced by partial fractions.

B An implementation-independent coefficient specification

For a fixed height m and requested order L , the following finite recipe fully specifies the coefficient calculation.

First compute $h_0, \dots, h_L$ from $\prod_{i < j} (1 - u(y_i + y_j)^2)^{-1}$ modulo u^{L+1} . Next compute the sign cumulants κ_r from

$$\log \cosh t = \sum_{r \geq 1} \kappa_r \frac{t^r}{r!}.$$

The even values begin $\kappa_2 = 1$, $\kappa_4 = -2$, $\kappa_6 = 16$, $\kappa_8 = -272$; the odd values vanish. The moment polynomials can be computed recursively as

$$\mu_0(z) = 1, \quad \mu_k(z) = \sum_{\substack{2 \leq r \leq k \\ r \text{ even}}} \binom{k-1}{r-1} \kappa_r z^{r/2-1} \mu_{k-r}(z) \quad (k > 0), \quad (\text{B.1})$$

with $\mu_k = 0$ for odd k . Apply (11.1) to each h_r , evaluate each monomial by (5.8), retain powers through z^{L-r} , and combine them as in (5.9). Finally apply the formal exponential (5.16). This recipe is sufficient to reproduce the printed coefficients independently of the shipped Python code.

For $m = 5$, the two outputs through order five are

$$(c_{5,j})_{j=0}^5 = \left(1, 10, \frac{225}{4}, \frac{1819}{8}, \frac{22045}{32}, 1353\right),$$

$$(b_{5,j})_{j=0}^5 = \left(1, \frac{48}{5}, \frac{5233}{100}, \frac{1028391}{5000}, \frac{60248377}{100000}, \frac{273939939}{250000}\right).$$

The first two nonconstant entries in each row also have the independent symbolic derivation in Section 6. Higher printed entries are exact evaluations of the proved finite formula, not inferred from observed numerical errors.

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